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GATE 2010 EE, 52^{nd} Question Analysis

Question 52

Statement for Linked Answer Questions: 52 & 53

The following Karnaugh map represents a function F.

F	00	01	11	10
X = 0	1	1	1	0
X = 1	0	0	1	0

52. A minimized form of the function F is

(A) $F = \bar{X}Y + YZ$ (B) $F = \bar{X}\bar{Y} + YZ$ (C) $F = \bar{X}\bar{Y} + Y\bar{Z}$ (D) $F = \bar{X}\bar{Y} + \bar{Y}Z$

Question Analysis:

- From above Kmap, the output F is as follows
- $F = \bar{X}.\bar{Y} + Y.Z$
- The Truth table for above expressions is as follows

The Truth Table

X	Y	Z	$F = \bar{X}.\bar{Y} + Y.Z$
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Hardware Implementation

The above problem is implemented and tested in hardware using Arduino UNO board. Here we used 3i/ps and LCD to display input and ouptu vlaue on LCD as per truth table and verified the expression.

Required Components & Pin Connections

S.No	**Component**
1	Arduino Uno Board
2	Breadboard
3	LCD (1)
4	Resistors: 220Ω (1)
5	Jumper Wires
6	USB Cable

Component	**Arduino Pin**
Input Z (5V or GND)	Digital 2
Input Y (5V or GND)	Digital 3
Input X (5V or GND)	Digital 4
Output RS (LCD RS)	Digital 8
Output EN (LCD EN)	Digital 9
Output D4 (LCD D4)	Digital 10
Output D5 (LCD D5)	Digital 11
Output D6 (LCD D6)	Digital 12
Output D7 (LCD D7)	Digital 13
GND	GND
VCC	5V

Logic Description

- Let initialize inputs $X = 0, Y = 0, Z = 0$
- The output expressions from KMAP and is as follows
- $F = \bar{X}.\bar{Y} + Y.Z$
- Change inputs and note Down the Experimental Truth table

Code Uploading Steps

1. Create a LCD folder and write Makefile
2. Write The code in main.c
3. Run the command "make". It will compile the code and creates .hex file
4. Copy the .hex file to ArduinoDriod folder
5. connect the Arduino UNO to mobile with OTG cable
6. Upload the hex file using "upload precomplied" option
7. Observe the ouput and verify the expression

Experimental Truth Table

X	Y	Z	F (Output)
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Conclusion

- From Experimental Truth Table and actual truth table $F = \bar{X}.\bar{Y} + Y.Z$
- This matches option (B) from the original GATE question.
- The hardware experiment confirms the circuit's theoretical logic.